Investigation case 2: SYNTHETIC demo: impossible position jump

This application is an educational maritime analytics portfolio project. Anomaly detections represent statistical or rule-based deviations, not evidence of illegal or hostile activity.

Analysis period: 2030-01-07 07:27 to 2030-01-07 07:28 UTC
Generated: 2026-10-19T18:57:18+00:00

Data sources

• synthetic_scenario / file n/a

Methods

• Detections from configurable rules and an Isolation Forest on vessel-day tracks.
• Evidence attached by an analyst; provenance captured when attached.

Case

Status: Explained. Created 2026-10-19 18:57 UTC by local-analyst.
One fix is ~15-30 km from its neighbours, implying a speed far above any vessel's.

Status history

old_status	new_status	changed_by	reason	changed_at
None	New	local-analyst	case opened	2026-10-19 18:57:17.983628+00:00
New	Under review	local-analyst	triage started	2026-10-19 18:57:17.999849+00:00
Under review	Explained	local-analyst	Single-fix outlier consistent with a pos	2026-10-19 18:57:18.003158+00:00

Detections under review

Event 3924 (position_jump): Consecutive positions are 18.46 km apart in 63.0 s (implied 569.6 kn). Possible benign explanations: GNSS or decoding error, or two transmitters configured with the same MMSI. To interpret this, review: vessel static data (name, call sign, dimensions) on each side of the jump. This is a rule-based deviation, not evidence of illegal or hostile activity.

Anomaly evidence

anomaly_event_id	anomaly_type	detector	anomaly_mmsi	event_time	score
3924.0	position_jump	rule	999010151	2030-01-07 07:27:49+00:00	None

Supporting observations

Positions

position_id	position_mmsi	base_datetime	lat	lon	sog
1709023.0	999010151	2030-01-07 07:27:49+00:00	40.52108206916396	-74.18777059829631	18.2
1709024.0	999010151	2030-01-07 07:28:52+00:00	40.42721232079113	-74.36779961127193	18.4

Analyst notes

Notes

added_at	added_by	note
2026-10-19 18:57:17.997162+00:00	local-analyst	Single isolated outlier fix; previous an

Evidence references

• anomaly_events.event_id=3924
• ais_positions.position_id=1709023
• ais_positions.position_id=1709024

Limitations

• AIS is self-reported and received by terrestrial stations; coverage varies and gaps are common. An observation gap is not evidence of deliberate transponder deactivation.
• NOAA Marine Cadastre data is downsampled to roughly one fix per minute per vessel.
• WPI ports are single reference points; 'near port' means within the configured radius of that point, not verified berthing.
• Rule thresholds are heuristics for this study area, not safety or regulatory standards.
• Isolation Forest scores rank statistical unusualness; they are not probabilities of any behaviour and say nothing about intent.